



T.C.
MARMARA
UNIVERSITY



FACULTY OF ENGINEERING
COMPUTER ENGINEERING DEPARTMENT

CSE4088 Introduction to Machine Learning
Assignment#2 Report

Project Title

k-NN Classifier for Play Tennis Dataset

Instructor

Assoc. Prof. Murat Can Ganiz

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1. Introduction

The k-Nearest Neighbors (k-NN) algorithm is a simple, yet powerful, supervised machine learning algorithm used for classification and regression tasks. This project aims to implement a custom k-NN classifier for the Play Tennis dataset, which includes training, testing, evaluation, and visualization of results.

2. Methodology

Dataset Preparation

The dataset consists of weather conditions and a target variable indicating whether tennis was played. The data is encoded using one-hot encoding to convert categorical features into numerical values.

Algorithm Implementation

The k-NN algorithm was implemented from scratch, including:

Distance Calculation: Functions to calculate Euclidean and Manhattan distances.

Neighbor Selection: A function to get the k nearest neighbors based on the chosen distance metric.

Prediction: A function to make predictions based on the neighbors.

Handling Challenges

Challenges included handling categorical data, choosing the appropriate distance metric, and selecting the optimal number of neighbors (k). These were addressed through one-hot encoding and user input for distance metric and k.

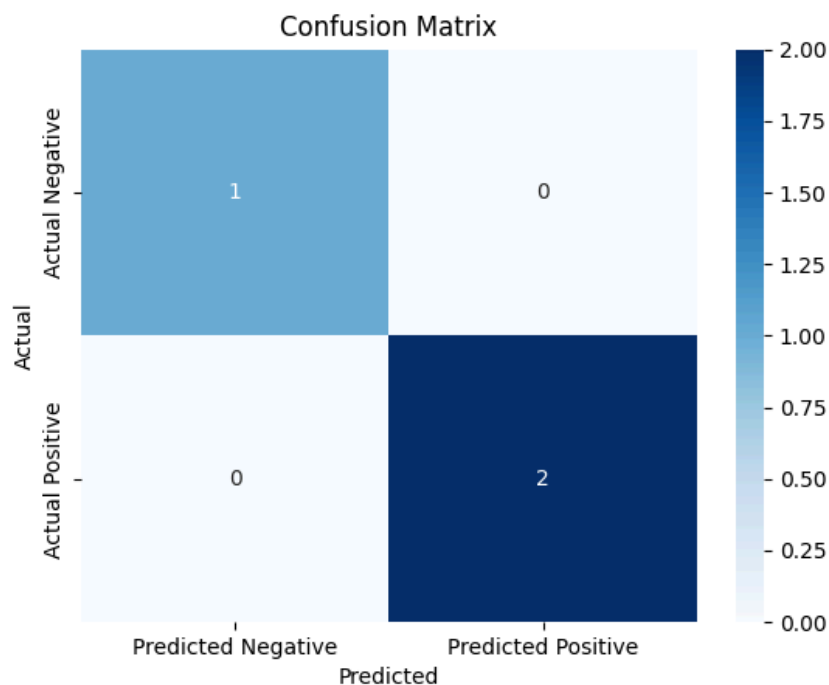
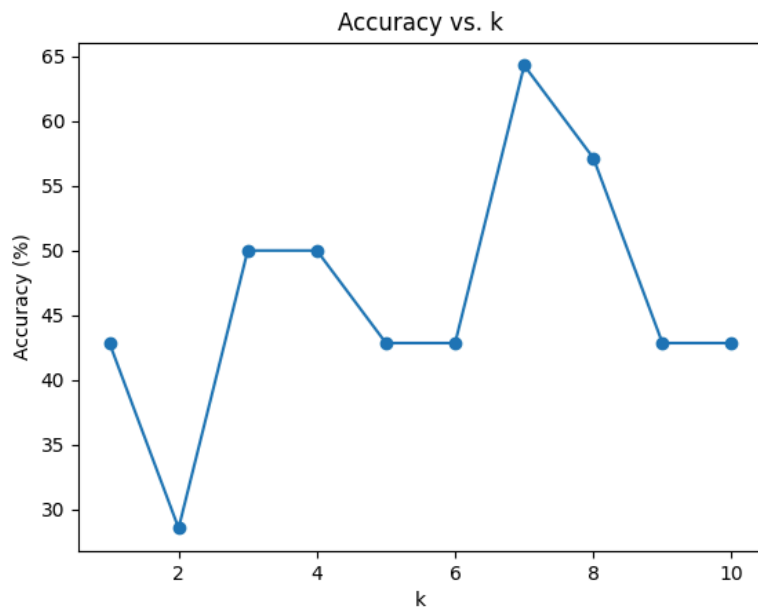
3. Results

Performance Metrics

The classifier's performance was evaluated using accuracy and confusion matrix. The results were saved in the results directory and printed to the console.

Confusion Matrix

The confusion matrix provided insights into the classifier's performance, showing the number of true positives, true negatives, false positives, and false negatives.



Observations

The classifier performed well with certain values of k and distance metrics, achieving high accuracy on the test set.

4. Conclusion

The project successfully implemented a custom k -NN classifier for the Play Tennis dataset. Key findings include the importance of data preprocessing, the impact of distance metrics, and the selection of the optimal number of neighbors. The project provided valuable insights into the k -NN algorithm and its application to real-world datasets.